



BANGALORE UNIVERSITY

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VASAVI JNANA PEETHA (Regd.)
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VASAVI JNANA PEETHA FIRST GRADE COLLEGE

Vijaynagar Bangalore -560040

(Recognized by Govt of Karnataka and Approved by Bangalore University)

PROJECT REPORT

This is to certify that this project work entitled

**“CROP RECOMMENDER SYSTEM USING MACHINE
LEARNING APPROACH”**

Submitted by

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*Student of 6th (Semester) final year in Practical fulfillment for the award of
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UNDER THE GUIDANCE OF

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ABSTRACT

Agriculture and its allied sectors are undoubtedly the largest sources of livelihood in rural India. Moreover, agriculture contributes significantly to the country's Gross Domestic Product (GDP). While the vast scale of India's agricultural sector is a national strength, it is regrettable that crop yield per hectare remains comparatively low by international standards. This disparity is considered one of the contributing factors to the high suicide rates among marginal farmers in India.

To address this issue, this paper proposes a viable and user-friendly crop yield prediction system tailored for Indian farmers. The proposed system is accessible via a mobile application, which leverages GPS to identify the user's location. Users input essential details such as land area and soil type. Based on this data, machine learning algorithms are employed to recommend the most profitable crops or predict the expected yield for a user-selected crop.

The system utilizes several machine learning techniques including Support Vector Machine (SVM), Artificial Neural Network (ANN), Random Forest (RF), Multivariate Linear Regression (MLR), and K-Nearest Neighbour (KNN). Among these, the Random Forest algorithm demonstrated the highest accuracy, achieving a prediction rate of 95%. Additionally, the system provides recommendations on the optimal timing for fertilizer application, further enhancing yield potential.

INTRODUCTION

What is Machine Learning?

Machine Learning is a system of computer algorithms that can learn from example through self-improvement without being explicitly coded by a programmer. Machine learning is a part of artificial Intelligence which combines data with statistical tools to predict an output which can be used to make actionable insights.

The breakthrough comes with the idea that a machine can singularly learn from the data (i.e., example) to produce accurate results. Machine learning is closely related to data mining and Bayesian predictive modeling. The machine receives data as input and uses an algorithm to formulate answers.

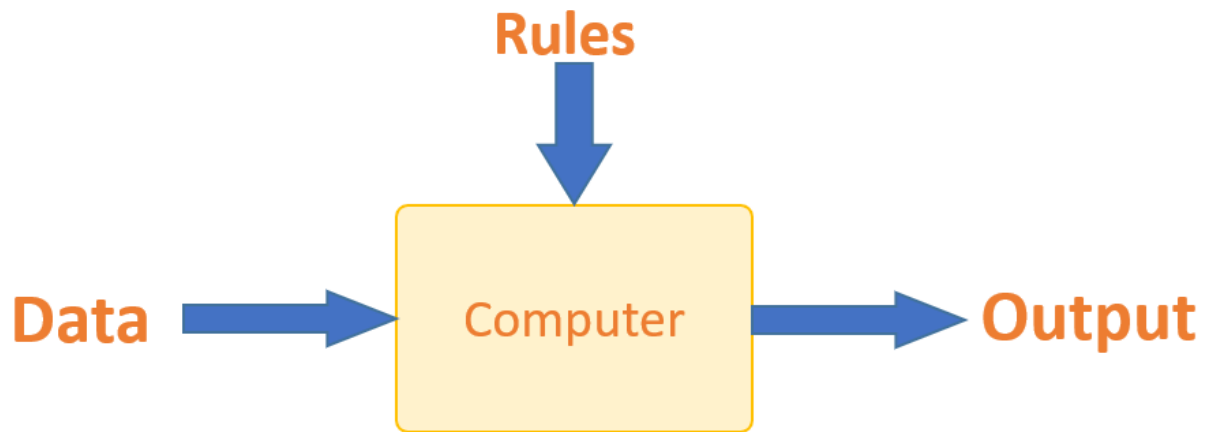
A typical machine learning tasks are to provide a recommendation. For those who have a Netflix account, all recommendations of movies or series are based on the user's historical data. Tech companies are using unsupervised learning to improve the user experience with personalizing recommendation.

Machine learning is also used for a variety of tasks like fraud detection, predictive maintenance, portfolio optimization, automatize task and so on.

Machine Learning vs. Traditional Programming

Traditional programming differs significantly from machine learning. In traditional programming, a programmer code all the rules in consultation with an expert in the industry for which software is being developed. Each rule is based on a logical foundation; the machine will execute an output following the logical statement. When the system grows complex, more rules need to be written. It can quickly become unsustainable to maintain.

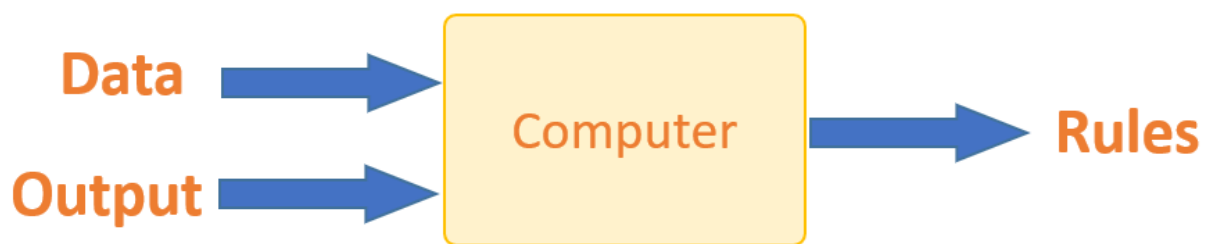
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Traditional Programming

Traditional Programming

Machine learning is supposed to overcome this issue. The machine learns how the input and output data are correlated and it writes a rule. The programmers do not need to write new rules each time there is new data. The algorithms adapt in response to new data and experiences to improve efficacy over time.



Machine Learning

Machine Learning

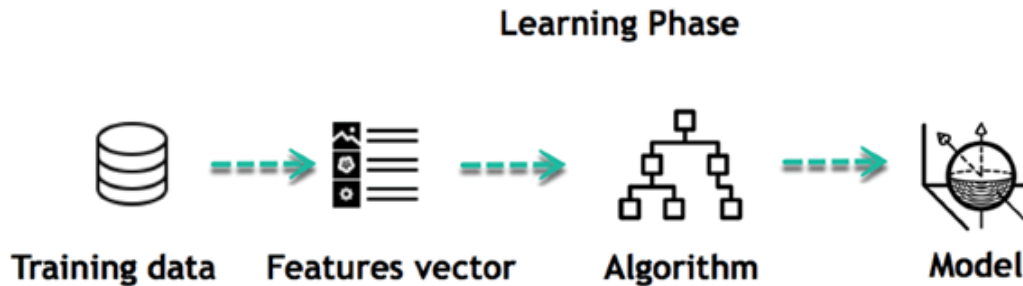
How does Machine Learning Work?

Machine learning is the brain where all the learning takes place. The way the machine learns is similar to the human being. Humans learn from experience. The more we know, the more easily we can predict. By analogy, when we face an unknown situation, the likelihood of success is lower than the known situation. Machines are trained the same. To make an accurate prediction, the machine sees an example. When we give the machine a similar example, it can figure out the outcome. However, like a human, if its feed a previously unseen example, the machine has difficulties to predict.

The core objective of machine learning is the **learning** and **inference**. First of all, the machine learns through the discovery of patterns. This discovery is made thanks to the **data**. One crucial part of the data scientist is to choose carefully which data to provide to the machine. The list

of attributes used to solve a problem is called a **feature vector**. You can think of a feature vector as a subset of data that is used to tackle a problem.

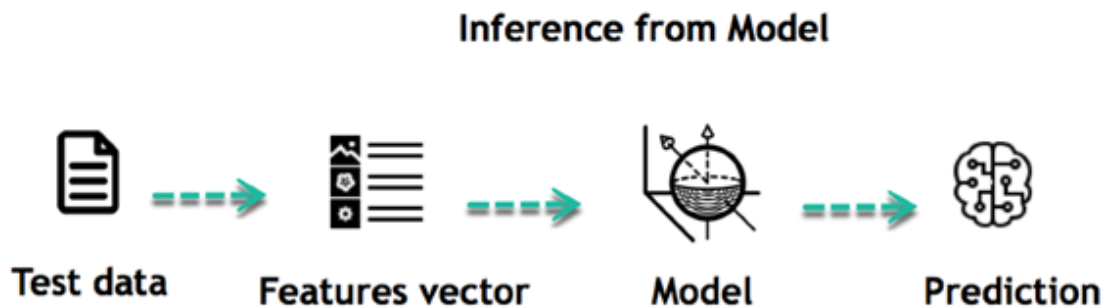
The machine uses some fancy algorithms to simplify the reality and transform this discovery into a **model**. Therefore, the learning stage is used to describe the data and summarize it into a model.



For instance, the machine is trying to understand the relationship between the wage of an individual and the likelihood to go to a fancy restaurant. It turns out the machine finds a positive relationship between wage and going to a high-end restaurant: This is the model

Inferring

When the model is built, it is possible to test how powerful it is on never-seen-before data. The new data are transformed into a features vector, go through the model and give a prediction. This is all the beautiful part of machine learning. There is no need to update the rules or train again the model. You can use the model previously trained to make inference on new data.



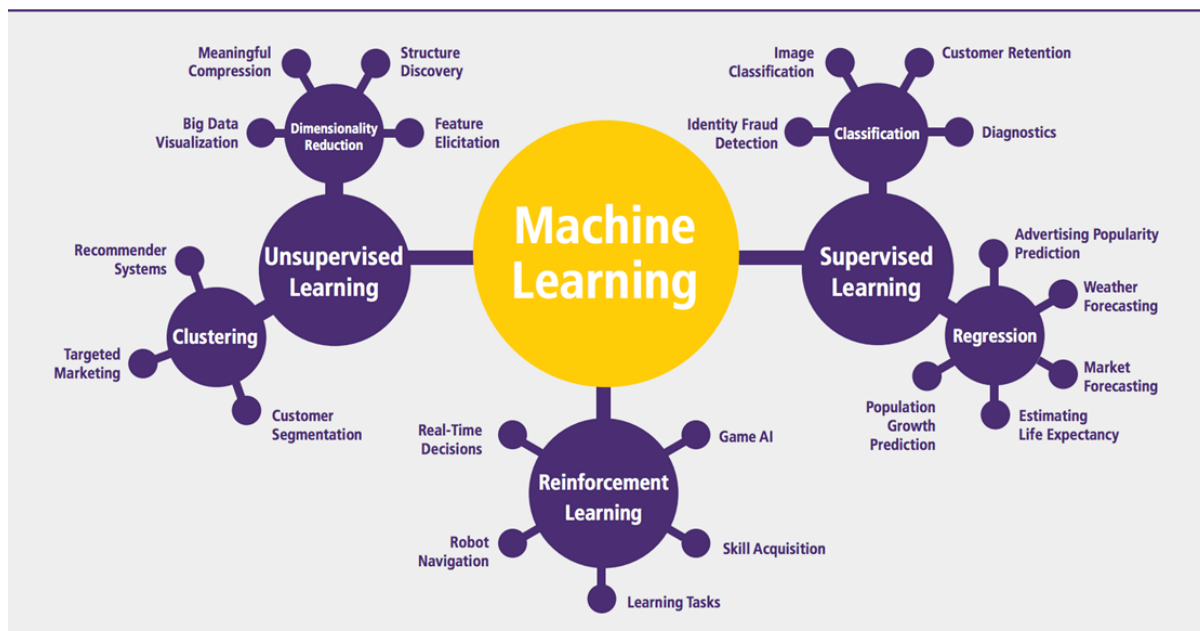
The life of Machine Learning programs is straightforward and can be summarized in the following points:

1. Define a question
2. Collect data
3. Visualize data
4. Train algorithm
5. Test the Algorithm

6. Collect feedback
7. Refine the algorithm
8. Loop 4-7 until the results are satisfying
9. Use the model to make a prediction

Once the algorithm gets good at drawing the right conclusions, it applies that knowledge to new sets of data.

Machine Learning Algorithms and Where they are Used?



Machine learning Algorithms

Machine learning can be grouped into two broad learning tasks: Supervised and Unsupervised. There are many other algorithms

Supervised learning

An algorithm uses training data and feedback from humans to learn the relationship of given inputs to a given output. For instance, a practitioner can use marketing expense and weather forecast as input data to predict the sales of cans.

You can use supervised learning when the output data is known. The algorithm will predict new data.

There are two categories of supervised learning:

- Classification task
- Regression task

SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ◆ ECONOMICAL FEASIBILITY
- ◆ TECHNICAL FEASIBILITY
- ◆ SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

SYSTEM ANALYSIS

EXISTING SYSTEM:

- ❖ Extensive work has been done, and many ML algorithms have been applied in the agriculture sector. The biggest challenge in agriculture is to increase farm production and offer it to the end-user with the best possible price and quality. It is also observed that at least 50% of the farm produce gets wasted, and it never reaches the end-user. The proposed model suggests the methods for minimizing farm produce wastage. One of the recent works, S. Pavani et.al. presented a model where the crop yield is predicted using KNN algorithms by making the clusters. It has been shown that KNN clustering proved much better than SVM or regression.
- ❖ Nishant et. al. predicted the crop yield for the specific year with the help of advanced regression techniques like Enet, Lasso and Kernel Ridge algorithms. The Stacking regression helped to enhance the accuracy of the algorithms.

DISADVANTAGES OF EXISTING SYSTEM:

- ❖ The main challenge faced in agriculture sector is the lack of knowledge about the changing variations in climate. Each crop has its own suitable climatic features. This can be handled with the help of precise farming techniques. The precision farming not only maintains the productivity of crops but also increases the yield rate of production.
- ❖ The existing system which recommends crop yield is either hardware-based being costly to maintain, or not easily accessible.
- ❖ Despite many solutions that have been recently proposed, there are still open challenges in creating a user-friendly application with respect to crop recommendation.

PROPOSED SYSTEM:

- ❖ In this project, we have proposed a model that addresses the existing issues. The novelty of the proposed system is to guide the farmers to maximize the crop yield as well as suggest the most profitable crop for the specific region.
- ❖ The proposed model provides crop selection based on economic and environmental conditions, and benefit to maximize the crop yield that will subsequently help to meet the increasing demand for the country's food supplies. The proposed model predicts the crop yield by studying factors such as rainfall, temperature, area, season, soil type etc. The system also helps to determine the best time to use fertilizers.
- ❖ The user provides an area under cultivation and soil type as inputs. According to the requirement, the model predicts the crop yield for a specific crop. The model also recommends the most profitable crop and suggests the right time to use the fertilizers.
- ❖ The main objective is to obtain a better variety of crops that can be grown over the season. The proposed system would help to minimize the difficulties faced by farmers in choosing a crop and maximize the yield.

ADVANTAGES OF PROPOSED SYSTEM:

- ❖ The proposed model predicts the crop yield for the data sets of the given region. Integrating agriculture and ML will contribute to more enhancements in the agriculture sector by increasing the yields and optimizing the resources involved. The data from previous years are the key elements in forecasting current performance.
- ❖ The proposed system uses recommender system to suggest the right time for using fertilizers.
- ❖ The methods in the proposed system includes increasing the yield of crops, real-time analysis of crops, selecting efficient parameters, making smarter decisions and getting better yield.

SYSTEM REQUIREMENTS

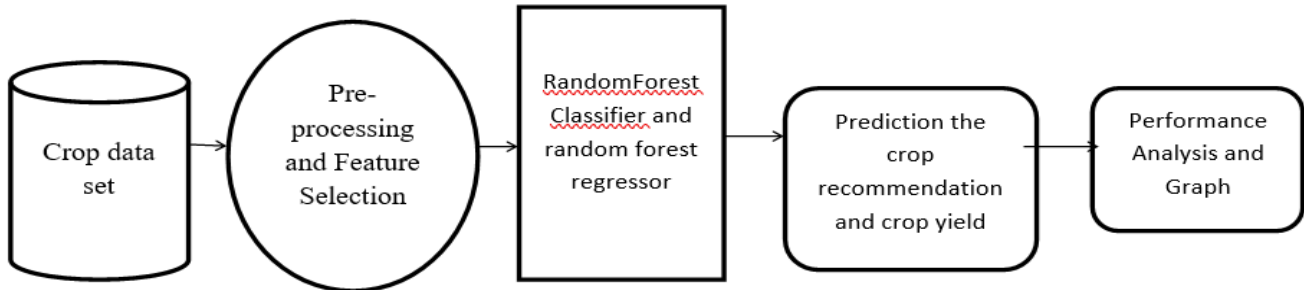
HARDWARE REQUIREMENTS:

- System : Pentium i3 Processor.
- Hard Disk : 500 GB.
- Monitor : 15’’ LED
- Input Devices : Keyboard, Mouse
- Ram : 4 GB

SOFTWARE REQUIREMENTS:

- Operating system : Windows 10.
- Coding Language : Python
- Web Framework : Flask

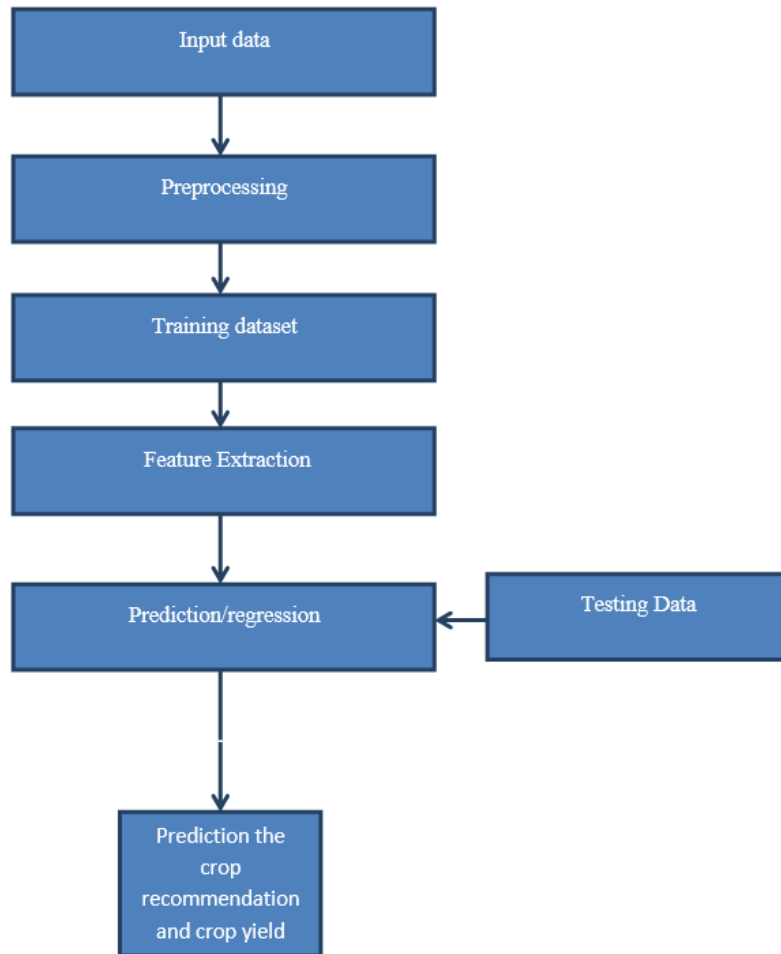
SYSTEM ARCHITECTURE:



DATA FLOW DIAGRAM:

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

CROP RECOMMENDER SYSTEM



UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

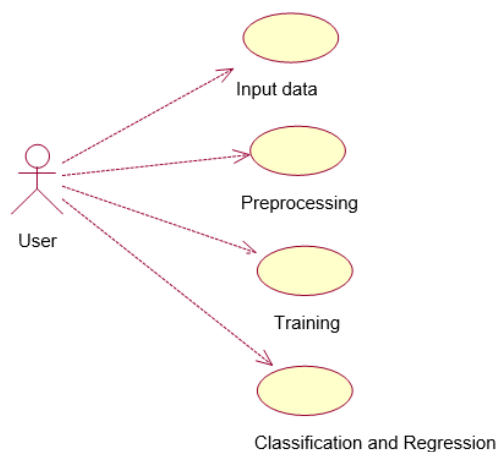
GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

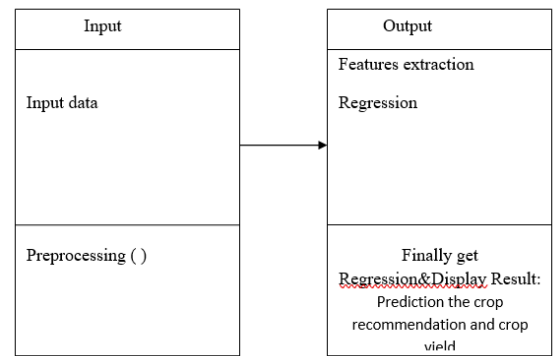
USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



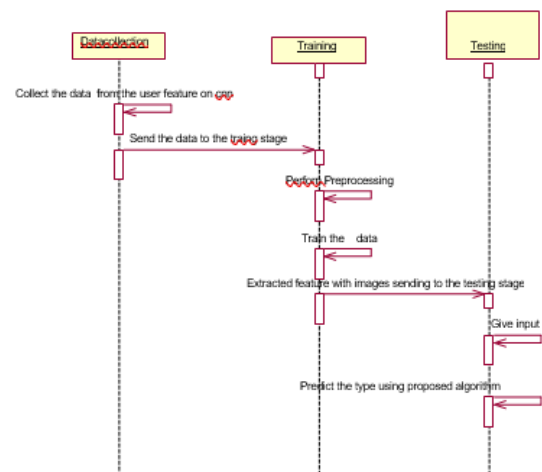
CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



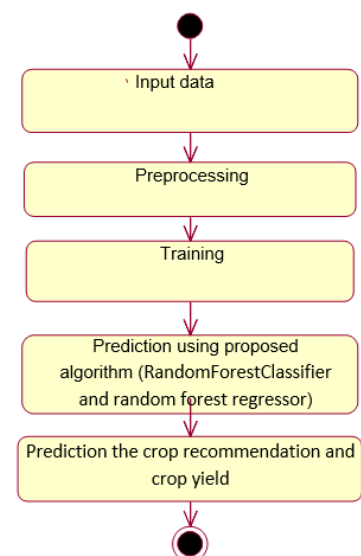
SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

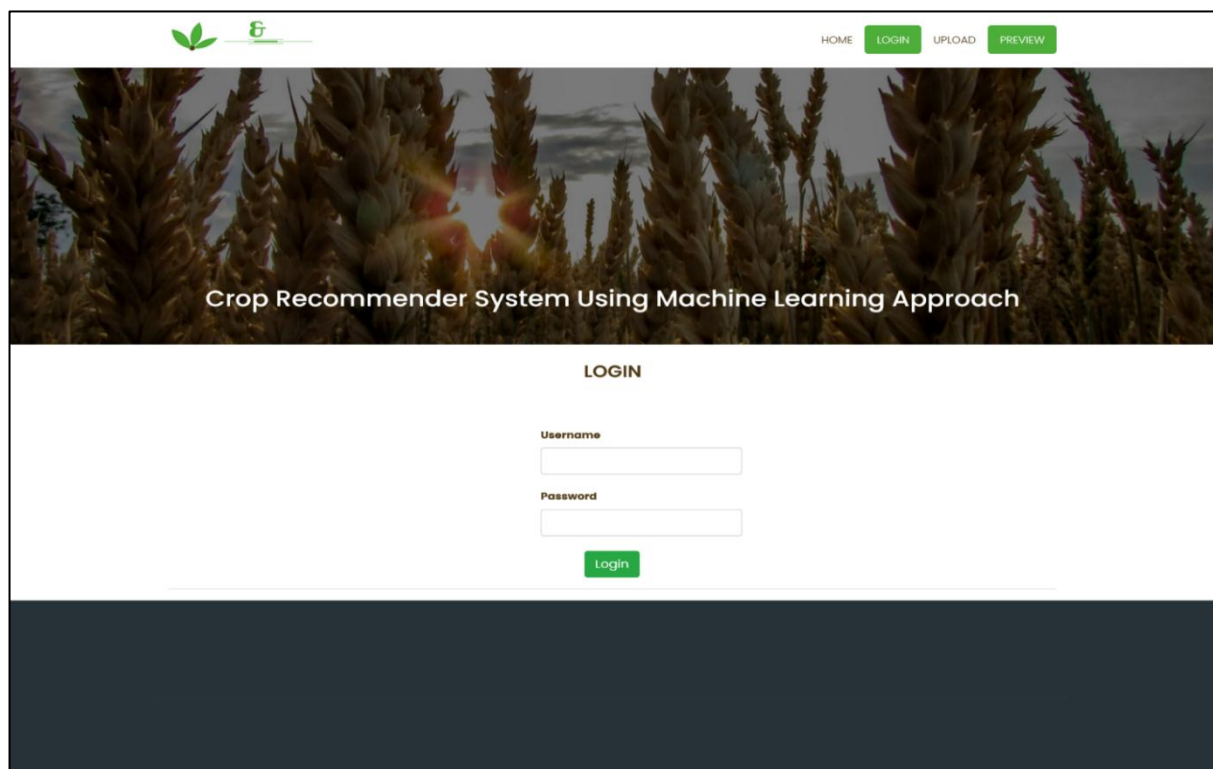
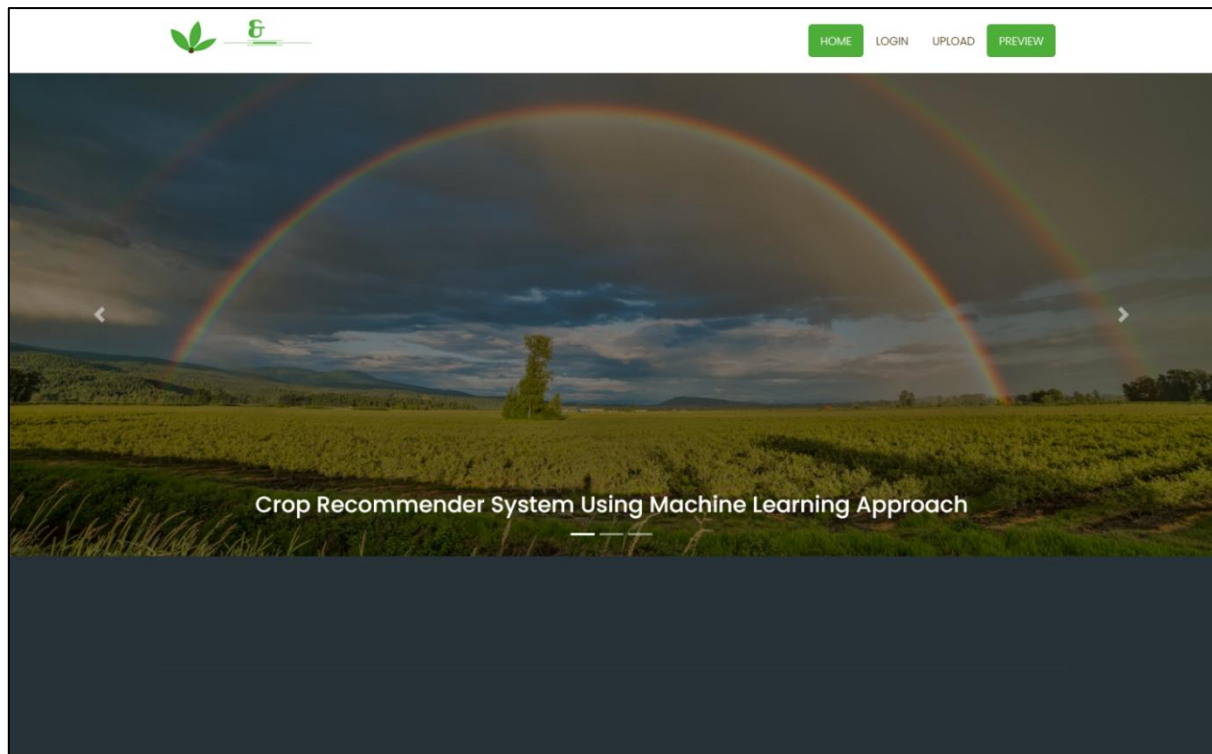


ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



SCREEN SHOTS:



CROP RECOMMENDER SYSTEM

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Crop Recommender System Using Machine Learning Approach

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HOME LOGIN UPLOAD **PREVIEW** YIELD PREDICTION

	State_Name	Season	Crop	Area	Production	soil_type
Id						
1	Andaman and Nicobar Islands	Kharif	Arecanut	1254.00	2.000000e+03	Laterite
2	Andaman and Nicobar Islands	Kharif	Arecanut	1254.00	2.061000e+03	Laterite
3	Andaman and Nicobar Islands	Whole Year	Arecanut	1258.00	2.083000e+03	Laterite
4	Andaman	Whole	Arecanut	1261.00	1.525000e+03	Laterite



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Click to Train | Test

CROP RECOMMENDER SYSTEM

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Crop Recommender System Using Machine Learning Approach

Crop Yield Prediction

Enter your Details

State_Name:

Crop:

Area :

soil_type:

Production would be **KGs**

Yield would be **KGs/acre**

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Crop Recommender System Using Machine Learning Approach

Crop Yield Prediction

Enter your Details

State_Name:

Crop:

Area :

soil_type:

Production would be **2024 KGs**

Yield would be **1614 KGs/acre**

CROP RECOMMENDER SYSTEM

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Crop Recommender System Using Machine Learning Approach

Crop Recommendation Prediction

Enter your Details

State_Name:

soil_type:

Area :

Crop Recommendation is :

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Crop Recommender System Using Machine Learning Approach

Crop Recommendation Prediction

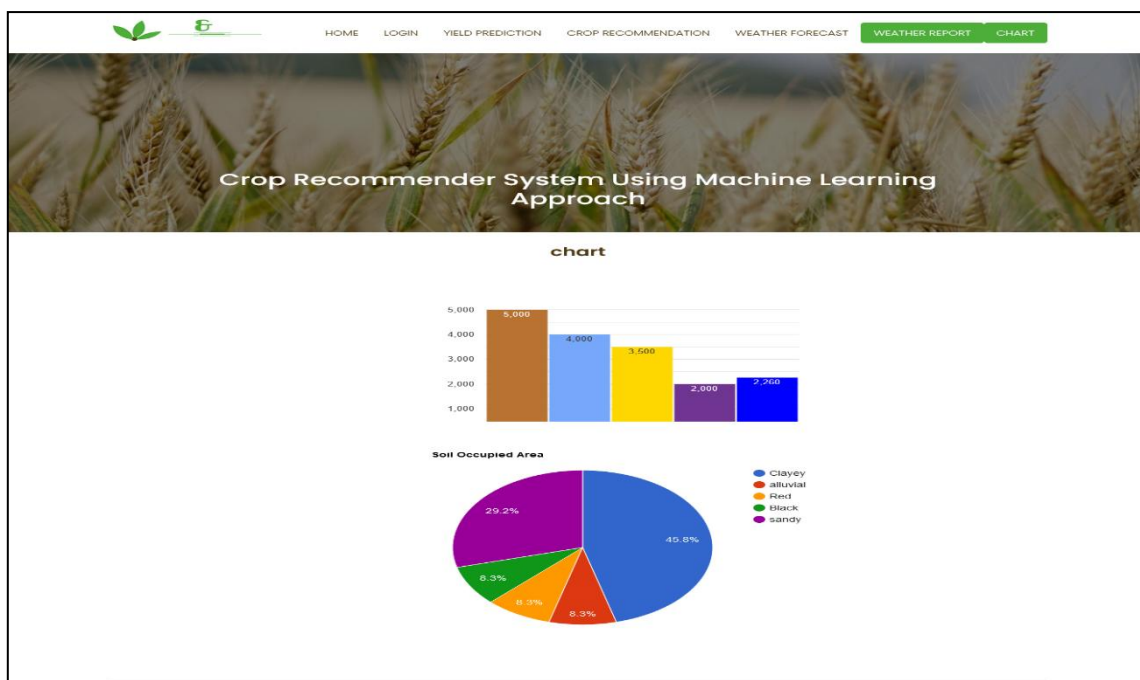
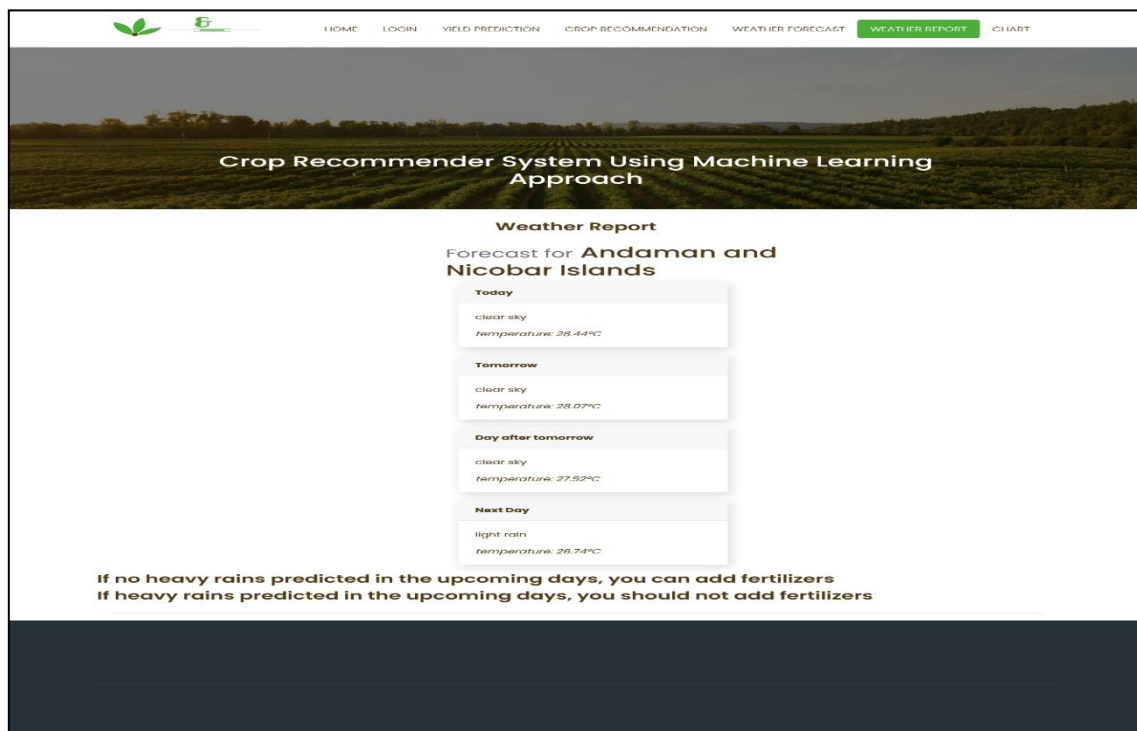
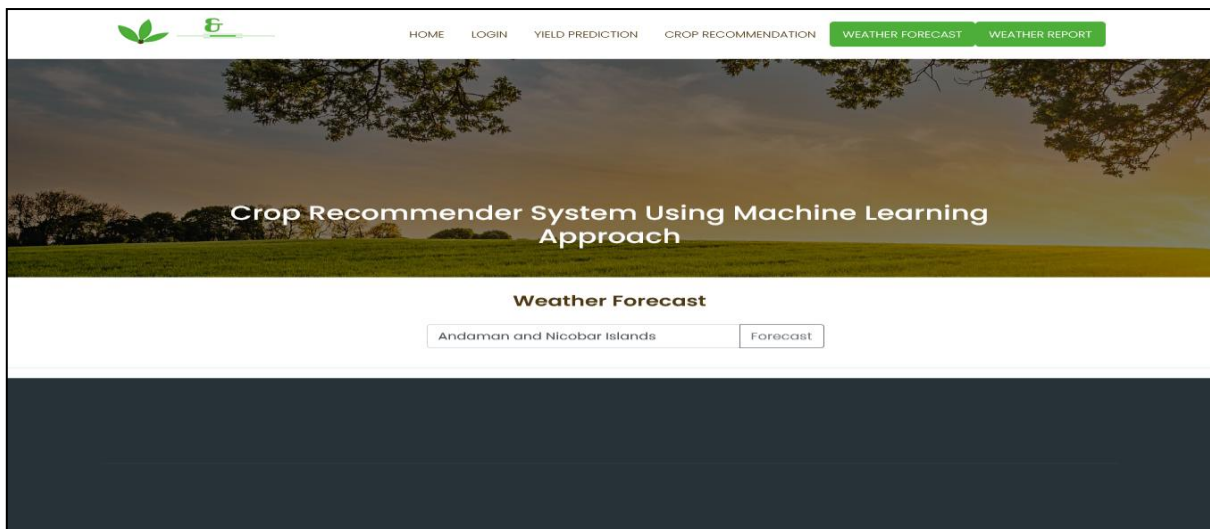
Enter your Details

State_Name:

soil_type:

Area :

Crop Recommendation is : Arecanut



CONCLUSION

This paper highlighted the limitations of current systems and their practical usage on yield prediction. Then walks through a viable yield prediction system to the farmers, a proposed system provides connectivity to farmers. The inbuilt predictor system helps the farmers to predict the yield of a given crop. The inbuilt recommender system allows a user exploration of the possible crops and their yield to take more educated decisions. For yield to accuracy, machine learning algorithm were implemented and tested on the given datasets. The proposed model also explored the timing of applying fertilizers and recommends appropriate duration.

Future Work:

The future work will be focused on updating the datasets from time to time to produce accurate predictions, and the processes can be automated. Another functionality to be implemented is to provide the correct type of fertilizer for the given crop and location. To implement this thorough study of available fertilizers and their relationship with soil and climate needs to be done. An analysis of available statistical data needs to be done.